

FRONTAL STORMS

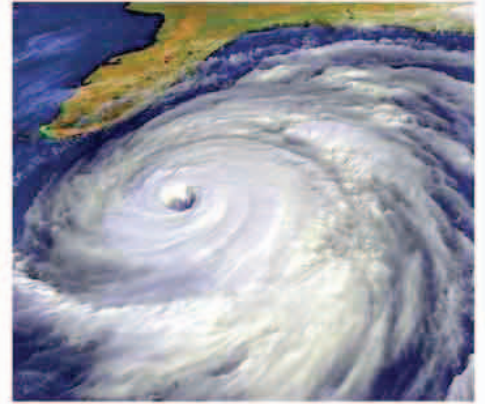
In the temperate regions just north and south of the tropics, warm tropical air is pushed up by cold polar air at an invisible boundary called the polar front. This helps generate cyclones over cool oceans such as the north Atlantic. The cyclones are swept east by westerly winds, and sometimes cause powerful storms like this one lashing a town on the Atlantic coast of Britain.



HURRICANES

Huge cyclones form over tropical oceans, where intense heat causes the build-up of colossal clouds around a zone of very low air pressure.

Air swirling into the low-pressure zone at high speed starts the clouds spinning, creating a tropical revolving hurricane. These storms, the most violent on Earth, are also known as typhoons or tropical cyclones.



▲ SPIRALING CLOUDS

This satellite view shows a hurricane going past Florida. Winds near the center of the hurricane can reach speeds of 220 mph (350 kph), causing immense destruction.

STORM SURGES

Extremely low air pressure at the center of a hurricane also makes the sea rise up like a tsunami wave—an effect called a storm surge. If the storm moves over land, it drags the storm surge with it, and if the surge is high enough, it can swamp coastal defenses and cause catastrophic flooding.



▲ FLOODED CITY

These houses in New Orleans were flooded by a storm surge swept ashore by Hurricane Katrina in August 2005.

